Lecture 1 - Learning from data

COMP3670/COMP6670 - Introduction to Machine Learning

Diego Marcondes

Semester 2 2026



Australian
National
University

Practical information

Poll Everywhere Quizzes

At the beginning of each lecture:

1. Use your device to log in to PollEverywhere (PE). Enter your ANU email (`uXXXXXXX@anu.edu.au`); PE supports ANU SSO.

   https://pollev.com/anuiml2026

2. Keep the webpage open. When the lecturer announces a quiz question, the webpage will be updated accordingly.



- In **COMP3670 P&C** or **COMP6670 P&C**, you can find the course overview description:

  Essential foundations for any machine learning (ML) application are a basic **statistical analysis of data** *to be processed, a solid understanding of the* **mathematical foundations underpinning ML**, *as well as* **basic classes of learning/adaptation concepts**. *The foundations are bundled in this single introductory course to ML in preparation for deeper explorations into the topic, but also as a standalone unit.*

COMP3670 / COMP6670

- This course covers selected topics from the following three mathematics areas, all highly relevant to ML:
  - calculus & optimization
  - probability & statistics
  - linear algebra
- The math topics will be integrated throughout the semester alongside ML concepts and algorithms, to highlight the direct relevance of math to ML.

Prerequisites

If you do not have a solid foundation in high-school level **mathematics** (algebra, coordinate geometry), functions & set theory, this course will be very challenging for you.

Teaching Team

- Course convenors and lecturers:
  - Diego Marcondes (Weeks 1–5.5, 10.5–12)
  - Yun Kuen (Marco) Cheung (Weeks 5.5–10.5)

- Tutors (in alphabetical order of last names):

Sunaina Ajmal
Oliver Bao
Subhransu Bhattacharjee
Ryan Guo
Isaac Leong
Saswat Panda
Geun Yun
Jinghan Zhang

Grading Scheme

There are five sets of assessment tasks.

1. **Three Assignments (7% each, 21% in total)**
2. **In-lecture Quizzes (4%)**: Starting from Week 2. Students must attend lectures to attempt the quizzes. Only your best 14 quizzes will be counted. No make-up or extension.
3. **Lab Exercises (4%)**: Students must attend lab sessions to attempt the exercises. Only your best 8 exercises will be counted. No make-up or extension.

   While the nominal total of the first three is 29%, their combined contribution to the final grade is capped at 22%.

4. **Mid-semester Test (23%)**
5. **Final Exam (55%)**

Grading Scheme

The final grade FG is calculated as:

$$FG = \min(7A_1 + 7A_2 + 7A_3 + 4Q + 4L,\ 22) + 23M + 55E$$

where A_1, A_2, A_3 are the percentages of available marks from the three assignments, and Q, L, M, E are the percentages for in-lecture quizzes, lab exercises, mid-semester test and final exam respectively.

Note: $A_1, A_2, A_3, Q, L, M, E \in [0, 1]$. For example, if you score 89% on the mid-semester test, plug in $M = 0.89$.

Course Activities

- Lectures:
  - When and where:
    - **Thursdays, 8:30–10:00am, Melville Hall, Room G01**
    - **Fridays, 8:30–10:00am, Copland Lecture Theatre**
  - From Week 2 onwards, there are quizzes in all regular lectures
  - There are quizzes in Week 1 too, but not counted toward the grade. They are for you to familiarise with the system.
  - Quiz questions appear at arbitrary times during a lecture.
- Labs (Weeks 2–11):
  - Sign up for a session:

    `https://mytimetable.anu.edu.au/even/student`
  - Each session includes 1.5 hours of "lab teaching" and 0.5 hours of "consultation" for you to ask tutors questions.

Canvas & Ed Forum

- Course materials will be uploaded to Canvas.
- Asking questions is a key strategy to learn successfully.
- Ed Forum is a math-friendly forum.
  Register at https://edstem.org/au/courses/37302/discussion
- Email convenors only for confidential (e.g., medical) matters.
  In such cases, use comp3670@anu.edu.au.
- For non-confidential but personal matters (e.g., timetabling), use Ed Forum with "Private" option.

Class Representatives

- **Class Representatives** are a core part of "student partnership", a principle to which the ANU has committed.
- A Class Representative is the official liaison between students and the course convener.
- The primary responsibility of a Class Representative is to provide constructive feedback on issues affecting the entire student cohort, or a subset of the student cohort.
- Class representatives also need to attend regular meetings arranged by School and College to provide reports on course feedback.

Benefits of being a Class Representative

- **Ensure students have a voice** to their course convener, lecturer, tutors, and College.
- **Develop skills sought by employers**, such as interpersonal, dispute resolution, leadership and communication skills.
- **Become empowered**, play an active role in determining the direction of your education.
- **Become more aware of issues influencing your University** and current issues in higher education.

Call for Class Representatives

- We are looking for 3 to 5 class representatives.
- Looking for diverse representation:
  - genders
  - domestic and international students
  - bachelor and master degree programs
- Apply with your CV via **this link**, before Monday Week 2.
  Write why you might be a good class representative.
  Describe your background or anything that might be relevant.

Academic Integrity

- ANU takes academic integrity seriously. **Plagiarism, collusion and contract cheating will not be tolerated.**
- This course introduces fundamental ML concepts that could be addressed by Generative AI tools (e.g., ChatGPT, Gemini, CoPilot). Hence, **the use of any Generative AI tool is not permitted in graded assessments within this course**.
- The quizzes and assignments prepare you for the test/exam. If you do not complete them yourself, it is unlikely that you will perform well in the test/exam. Remember the final exam has a hurdle.

Academic Integrity

- Examples of plagiarism:
  - Your assignment solutions are highly similar to other students'.
  - Your solutions are highly similar to a webpage (and potentially similar to other students' who also referred the same webpage).
  - You fail to cite the external reference where you get your idea from (but your solution is sufficiently different from the external reference).
  - You fail to cite your peer who discussed with you (but your solutions are sufficiently different from your peers').
  - You reuse your own codes/assignments from another course, or from your previous attempt of the same course. This is known as self-plagiarism.

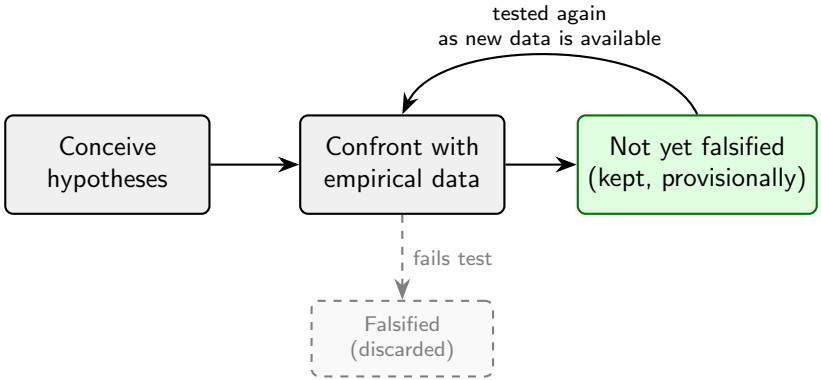
Academic Integrity

- Example of collusion:
  - You discuss an individual assignment with a friend/classmate, and acknowledge their contribution to your assignment.
- Examples of contract cheating:
  - You use a solution taught by a tutoring company.
  - You use a social network to find and pay someone to do your assignment.
- For more detailed guidance on academic integrity, see
  https://www.anu.edu.au/students/academic-skills/academic-integrity

Science and inductive reasoning

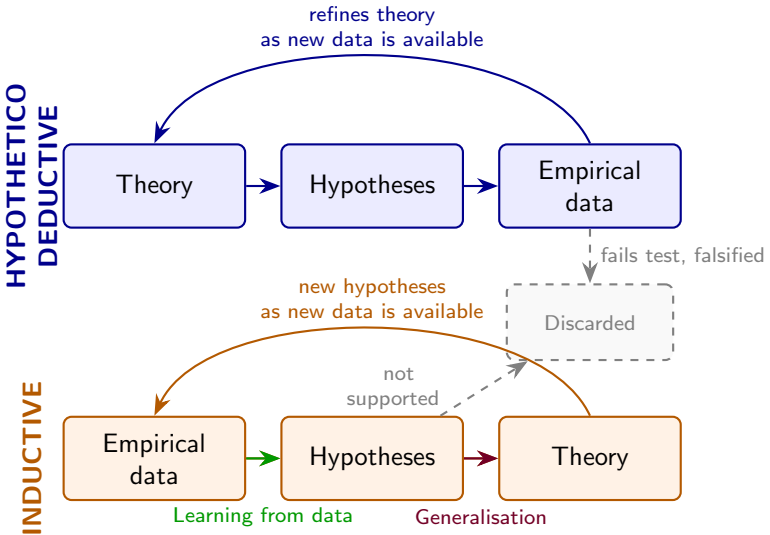
What is science?

- Modern science can be generally described as a three-step process:



A hypothesis, that is **a proposed explanation for a phenomenon**, *is never proven true: it is only, so far, not falsified.*

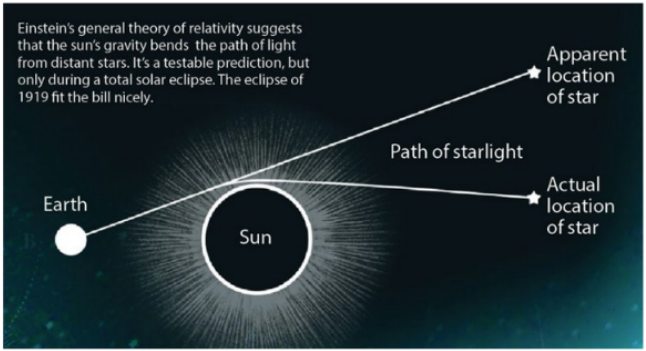
Two approaches to scientific reasoning



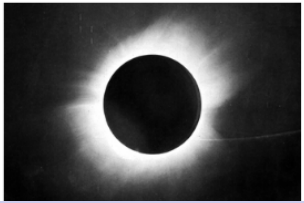
Hypothetico-deductive reasoning

- We start with an established **theory**
  - **Theory**: Einstein's general relativity (1915)
- The theory implies certain **hypotheses**, which are predictions that must hold if the theory is correct
  - **Hypothesis implied by the theory**: light from distant stars should bend as it passes near the sun
- We confront these hypotheses with empirical evidence, attempting to **falsify** them
  - **Test**: Eddington's 1919 solar eclipse expedition measured the apparent position of stars near the sun
- If a hypothesis is falsified, the theory must be revised or discarded, otherwise the theory stands for now
  - **Result**: the predicted bending was observed, hence the hypothesis was **not falsified** and the theory stood

Hypothetico-deductive reasoning



Source: Diagram & 1919 eclipse https://earthsky.org/human-world/may-29-1919-solar-eclipse-einstein-relativity/ and The New York Times of November 10, 1919 en.wikipedia.org/wiki/Eddington_experiment



LIGHTS ALL ASKEW IN THE HEAVENS

Men of Science More or Less Agog Over Results of Eclipse Observations.

EINSTEIN THEORY TRIUMPHS

Stars Not Where They Seemed or Were Calculated to be, but Nobody Need Worry.

A BOOK FOR 12 WISE MEN

No More in All the World Could Comprehend It, Said Einstein When His Daring Publishers Accepted It.

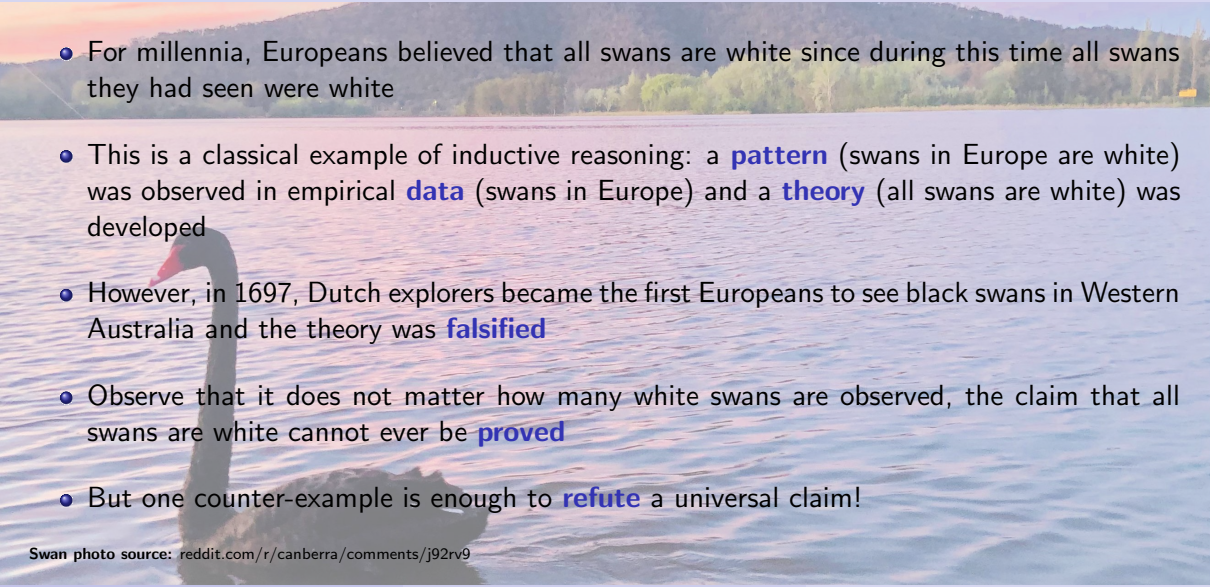
Inductive reasoning

- We start with **observations** and no theory is assumed beforehand
  - **Data collection**: The astronomer Tycho Brahe (1546-1601) spent decades recording precise positions of the planets
- We look for a **pattern** or hypothesis that fits these observations
  - **Find a pattern**: Johannes Kepler (1571-1630) studied these records and inferred a pattern: planetary orbits are **ellipses**, not circles
- We confront the hypothesis with further empirical evidence and, as more evidence fails to falsify it, a "**theory**" is gradually established
  - **A law that works**: Kepler had **no theory** that correctly explained *why* orbits should be elliptical, but the law emerged purely from the data

Is induction reliable?

- Decades later, Newton's theory of gravitation **deductively explained** why orbits must be ellipses
- So the two modes of reasoning are not necessarily opposed and they can **complement** each other
- While induction proposes a pattern, deduction may later explain **why** it holds
- Much of science alternates between the two, however:
  - Deduction is **logically safe**: if the premises are true, the conclusion must be true
  - Induction is **not**: no finite number of observations can logically **guarantee** a universal claim
- This is known as **Hume's problem of induction**
- Let us see why this matters with a classic example

The problem of induction: an example



- For millennia, Europeans believed that all swans are white since during this time all swans they had seen were white
- This is a classical example of inductive reasoning: a **pattern** (swans in Europe are white) was observed in empirical **data** (swans in Europe) and a **theory** (all swans are white) was developed
- However, in 1697, Dutch explorers became the first Europeans to see black swans in Western Australia and the theory was **falsified**
- Observe that it does not matter how many white swans are observed, the claim that all swans are white cannot ever be **proved**
- But one counter-example is enough to **refute** a universal claim!

Swan photo source: reddit.com/r/canberra/comments/j92rv9

Popper's answer: falsifiability

- If we can never **prove** a general claim, how do we tell science from non-science?
- This is the **demarcation problem**
- Karl Popper's answer: a claim is **scientific** if and only if it can, in principle, be **falsified** by some possible observation
- A claim compatible with **every** possible observation tells us nothing about the world
- Within the induction principle: a collection of hypothesis that **fit** any data we can possibly observe cannot provide a **real** pattern that **generalises** to new data

Unfalsifiability: an example

- Consider the claim: *"a secret society completely controls the global economy and hides all traces of its existence"*
- This claim cannot be falsified:
  - If one finds evidence of its corruption, it proves the secret society exists;
  - If one looks closely and finds absolutely zero evidence of a conspiracy, the theorist will claim that the lack of evidence is just proof of how powerful, clever, and thorough the secret society is at covering its tracks.
- No observation could ever refute this claim
- By Popper's criterion, this is **not** a scientific claim, since it is unfalsifiable

Machine Learning is inductive learning

- **Machine Learning**: algorithms that learn patterns from data, and use them to make predictions on **new, unseen** data
- This is inductive reasoning, made precise and computational: **learn a hypothesis** from data and then **generalise** to new data from the same phenomenon
- We observe a **finite data sample** and want a rule that holds **beyond** it

Falsifiability in Machine Learning

- Since ML is inductive, Hume's problem of induction is, therefore, **unavoidable** in Machine Learning
- Suppose a model (i.e., collection of hypotheses) is **so flexible** that it can perfectly fit **any** possible dataset, no matter what the data looks like
- Such a model can never be falsified by data, much like the "secret society" example
- It fits everything, and therefore explains nothing
- This phenomenon is related to an important concept in Machine Learning: **overfitting**
- We will return to this idea, precisely, throughout the course

Introduction to Machine Learning outline

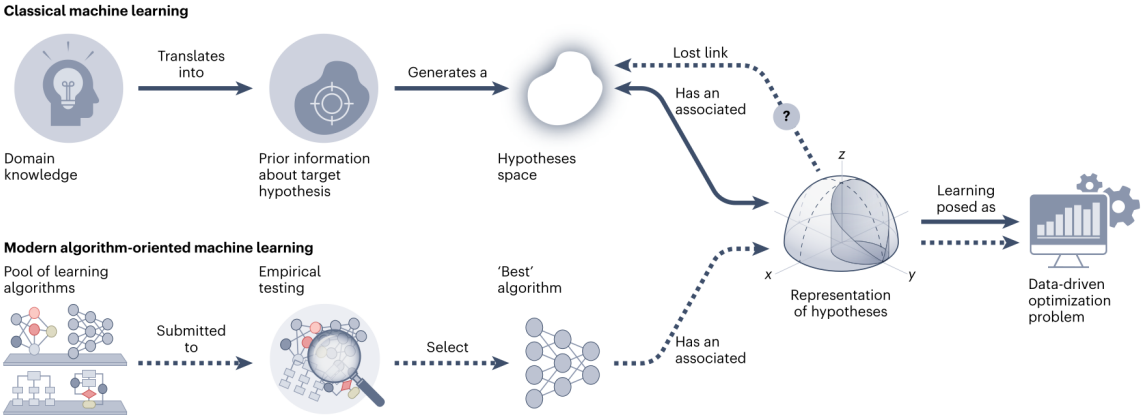
In this course, we will learn the fundamental aspects of Machine Learning:

- Basic **concepts and types** of Machine Learning
- Methods for **supervised and unsupervised learning**
- **Probability and statistics**: how can we mathematically define "generalisation" and how can we control it?
- **Optimisation**: how can we learn from data, that is, "train a model"?
- The basic mathematics necessary for machine learning: **linear algebra and calculus**

Main course takeaway

Machine learning is **NOT** merely an algorithmic way of finding patterns in data: it is based on a **statistical model** within inductive reasoning and the algorithm is only the **tool** used to fit this model to data, not the model itself.

Learning from data pipeline



Reference: Marcondes, D.; Simonis, A.; Barrera, J. Back to basics to open the black box. Nature Machine Intelligence. 2024

Types of Machine Learning

Information available in ML

- Every Machine Learning problem involves some **inputs** (features, measurements):
  - **House price prediction**: size, number of rooms, location
  - **Medical diagnosis**: patient age, blood pressure, cholesterol level
  - **Image classification**: pixel intensities of an image
  - **Spam detection**: frequency of certain words in an email
  - **Credit scoring**: income, credit history, loan amount requested
- What differs between the types of ML is **what else** is available:
  - Are **labels or outcomes** observed for each input?
  - Is there a **feedback signal** instead of a fixed dataset?
- This gives us three broad types of Machine Learning: **supervised**, **unsupervised** and **reinforcement learning**

Supervised learning

- We observe **labelled** examples: $(\mathbf{x}_1, y_1), \ldots, (\mathbf{x}_N, y_N)$
  - **Notation**: we denote x for inputs and y for labels, outcomes or outputs. When the quantity is actually a **vector** $(x_1, \ldots, x_n)$ of values instead of a unique value, we denote it with boldface $\mathbf{x}$. The set of all possible values of $\mathbf{x}$ is denoted by $\mathcal{X}$ and of y by $\mathcal{Y}$. The dataset is denoted by $\mathcal{D} = \{(\mathbf{x}_1, y_1), \ldots, (\mathbf{x}_N, y_N)\}$.

- Each $\mathbf{x}_i$ is a **feature vector**, and each y_i is the known **outcome** for that input. We have a dataset with N examples of input and outcome

- **Task**: learn a function (rule) $h : \mathcal{X} \to \mathcal{Y}$ such that $h(\mathbf{x}_i) \approx y_i$ for all $i = 1, \ldots, N$ that also works on **new**, **unseen** inputs (i.e., **generalises**)

- This is the type of Machine Learning we will focus on in the following weeks

Supervised learning: classification and regression

- **Classification**: the outcome y takes values in a **finite** set
  - *e.g.*, $\mathcal{Y} = \{\text{spam}, \text{not spam}\}$, or $\mathcal{Y} = \{\text{approve}, \text{deny}\}$, or $\mathcal{Y} = \{\text{dog}, \text{cat}, \text{horse}\}$
  - Since the outcome must be a number we take $\mathcal{Y} = \{0, 1\}$, or $\mathcal{Y} = \{-1, +1\}$, or $\mathcal{Y} = \{0, 1, \ldots, c-1\}$ when there are c classes, for example
- **Regression**: y takes **continuous** values
  - *e.g.*, $y =$ house price, or $y =$ tomorrow's temperature
  - In this case,
    - $\mathcal{Y} = \mathbb{R} =$ All real numbers
    - $\mathcal{Y} = \mathbb{R}_+ =$ All positive real numbers
    - $\mathcal{Y} = [a, b] =$ All real numbers between a and b
- Both are supervised: in both, labels are observed in the dataset

Examples of classification and regression problems

- **Classification examples**:
  - **Spam detection**:

$$\mathbf{x} = \text{frequency of words in an email}, y \in \{\text{spam}, \text{not spam}\}$$

  - **Medical diagnosis**:

$$\mathbf{x} = \text{patient's symptoms, test results}, y \in \{\text{disease present}, \text{disease absent}\}$$

  - **Image classification**:

$$\mathbf{x} = \text{pixel intensities of an image}, y \in \{\text{dog}, \text{cat}, \text{horse}\}$$

- **Regression examples**:
  - **House price prediction**:

$$\mathbf{x} = \text{size, number of rooms, location}, y = \text{price}$$

  - **Weather forecasting**:

$$\mathbf{x} = \text{today's temperature, humidity, pressure}, y = \text{tomorrow's temperature}$$

  - **Life expectancy**:

$$\mathbf{x} = \text{lifestyle and health indicators}, y = \text{expected lifespan (years)}$$

Unsupervised learning

- We observe **only** inputs: $\mathbf{x}_1, \ldots, \mathbf{x}_N$, with no labels at all
- **Task**: discover **structure** hidden in the data
- Common tasks:
  - **Clustering**: group similar points
    - ⋆ **Customer segmentation**: Discover groups of customers with similar behaviour, with no predefined categories

      $\mathbf{x}$ = purchase history, browsing behaviour, demographics
  - **Dimensionality reduction**: compress data while preserving structure
    - ⋆ **Genomics**: Compress into a small number of variables that capture the main sources of variation

      $\mathbf{x}$ = expression levels of thousands of genes per patient
  - **Density estimation**: model the distribution of $\mathbf{x}$
    - ⋆ **Fraud detection**: Model the distribution of normal transactions and flag those in low-probability regions

      $\mathbf{x}$ = transaction amount, location, time, merchant
- Without labels, even defining a "good" solution is part of the problem

Reinforcement learning

- In supervised learning, each training example provides the **correct output**; in unsupervised learning, **no feedback** is given at all
- In reinforcement learning, for each example we have something in between: a **grade** on how good the model output was, but not what the correct output was supposed to be
- An **agent** observes a **state** $\mathbf{x}$, takes an **action**, and receives a **reward** signal (grade)
- The agent must discover on its own **which actions** lead to a good or bad outcome
- **Task**: learn a **policy** $h : \mathcal{X} \to \mathcal{A}$ mapping states to actions that maximises cumulative reward

Examples of reinforcement learning problems

- **Game playing (Chess)**:
  - **State x**: current configuration of the board
  - **Action**: which piece to move, and where among the possible moves
  - **Reward**: $+1$ for winning, -1 for losing, 0 if the game has not ended after the move
  - A game has dozens of moves, but reward only comes at the end: the agent must figure out *which moves* were responsible for winning or losing
- **Personalised recommendations**:
  - **State x**: user profile, browsing history, time of day
  - **Action**: which item, video, or article to recommend
  - **Reward**: user clicks, watch time, purchase, i.e., a grade on the quality of the recommendation

Comparing the three types

- **Supervised**: inputs *and* labels observed; goal is prediction
- **Unsupervised**: only inputs observed; goal is structure discovery
- **Reinforcement**: no fixed data; agent generates it through interaction and feedback
- All three, however, share the same underlying inductive challenge: generalising **beyond** what has been observed in the data
- In this course we will focus on **supervised** and **unsupervised** learning

The (Supervised) Learning Framework

A concrete problem: credit approval

Example

A bank wants to automatically approve or deny credit card applications, based on information about each applicant.

- Each applicant is described by a **feature vector**

$$\mathbf{x} = (x_1, x_2, \ldots, x_n)$$

- For example:
  - x_1 = annual income
  - x_2 = years employed
  - x_3 = outstanding debt
  - ...

Feature	**Applicant 1**	**Applicant 2**
Annual income	\$52,000	\$28,000
Years employed	5	1
Outstanding debt	\$3,200	\$15,400
Outcome y	**Repaid (+1)**	**Defaulted (-1)**

Task: learn from data a rule to predict the outcome of **future** applicants

The target function

- In any supervised learning problem, we assume there exists an **ideal rule** $f : \mathcal{X} \to \mathcal{Y}$ that **best** maps every input to its output
  - *e.g.*, in credit approval: the rule that performs as well as possible, with loan application information, on determining whether an applicant will repay or default
- The target function f is **unknown**: it is precisely what we are trying to discover from data
- What we observe is a **finite dataset** of input-output pairs that **reflects** f, but possibly with noise: measurements can be imperfect, outcomes uncertain
- **Goal of learning**: use these observations to find a rule h that approximates f well enough to make reliable predictions on **new, unseen** inputs (i.e., generalise!)
- What "target function" and "approximates f well" mean *precisely* requires the language of **probability and statistics**: this will be made exact in the next lecture

The hypothesis set: all possible rules

- Recall the first step of inductive reasoning: before confronting hypotheses with data, we must **conceive** them
- In the credit approval problem, a **hypothesis** h is any candidate rule that reads the features and outputs a decision:

$$h : \mathcal{X} \to \{+1, -1\}$$

- The **hypothesis set** $\mathcal{H}$ is the collection of *all* such candidate rules **we are willing** to consider before seeing any data
- For example, some candidate hypotheses for credit approval:
  - Approve if annual income $>$ \$40,000
  - Approve if income is high *and* outstanding debt is low
  - Approve if a weighted combination of all features exceeds a threshold
- In **every** supervised machine learning method there is an underlying hypothesis set, even if in practice one does not *"care"* about it

The learning algorithm

- Once $\mathcal{H}$ is fixed, we need a principled way to **select** one hypothesis from it based on data
- **The key idea**: each hypothesis $h \in \mathcal{H}$ makes predictions on the training data, and we can **measure how wrong** those predictions are
  - *e.g.*, in credit approval: how many past applicants did h classify incorrectly?
- A **learning algorithm** $\mathcal{A}$ searches through $\mathcal{H}$ and selects the hypothesis that **minimises this error on the training sample**
- The selected hypothesis is the **learned hypothesis** $\hat{h}$ that we will use as an approximation of the target f
- Crucially, minimising error on the training sample is **not** the goal: the goal is to have small error on **future, unseen** inputs
- Whether these two are related is the **central** question of **Statistical Learning Theory** (more detail on the next lecture)

In mathematical notation

- **Training sample**: $\mathcal{D} = \{(\mathbf{x}_1, y_1), \ldots, (\mathbf{x}_N, y_N)\}$, with $\mathbf{x}_i \in \mathcal{X}$ and $y_i \in \mathcal{Y}$
- **Hypothesis set**: $\mathcal{H}$ a set of candidate functions $h : \mathcal{X} \to \mathcal{Y}$
- **In-sample error**: $E_{\text{in}}(h)$ a measure of how wrong h is on $\mathcal{D}$
  - *e.g.*, in credit approval: the fraction of past applicants that h classifies incorrectly in the dataset

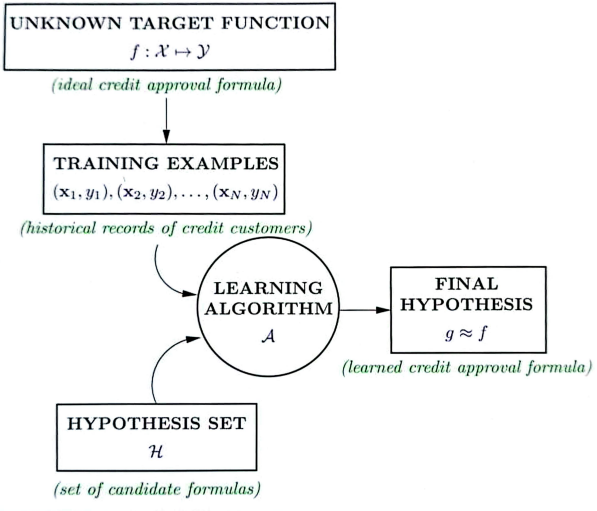
$$E_{\text{in}}(h) = \frac{1}{N} \sum_{i=1}^{N} \mathbb{1}\{h(\mathbf{x}_i) \neq y_i\}$$

  in which $\mathbb{1}\{\cdot\} = 1$ if the expression inside it is True and $\mathbb{1}\{\cdot\} = 0$ if it is False

- **Learning from data**: solving the optimisation problem

$$\hat{h} = \underset{h \in \mathcal{H}}{\arg\min} \; E_{\text{in}}(h)$$

The learning framework



Source: Abu-Mostafa, Y., Magdon-Ismail, M., & Lin, H.-T. (2012). *Learning from Data*. AMLBook. Figure 1.2.

Choosing the hypothesis set $\mathcal{H}$

- Choosing $\mathcal{H}$ is a crucial **design decision**, made *before* learning begins
- $\mathcal{H}$ may encode our **prior knowledge** about the likely form of f
- There is, however, a fundamental **trade-off** in this choice:
  - If $\mathcal{H}$ is **too simple**, it may not contain any function close to f
    - ⋆ No matter how much data we collect, or how well we search $\mathcal{H}$, we cannot do better than the best hypothesis $\mathcal{H}$ contains
    - ⋆ This limitation is called the **approximation error**
  - If $\mathcal{H}$ is **too complex**, it becomes easy to fit *any* training data, including *noise*
    - ⋆ Recall our earlier discussion: a model so flexible it can fit anything is **unfalsifiable**
    - ⋆ Fitting the training data perfectly may no longer tell us anything reliable about f, especially if the process has some **randomness**
    - ⋆ This is the seed of **overfitting**
- This is known as the **bias-variance trade-off** and will be studied further in future lectures

A Concrete Example: The Perceptron

From framework to a concrete hypothesis set

- So far, $\mathcal{H}$ has been abstract
- We now fix a specific, simple $\mathcal{H}$: the set of **linear threshold functions**
- The resulting learning problem is called the **perceptron**
- It is one of the oldest and simplest Machine Learning models, and an excellent way to see the supervised ML framework in action

The perceptron hypothesis set

- **Binary classification**: $\mathcal{Y} = \{+1, -1\}$, features $\mathbf{x} = (x_1, \ldots, x_n) \in \mathcal{X} = \mathbb{R}^n$ = All vectors of n real numbers
- Each hypothesis in $\mathcal{H}$ has the form

$$h_{\mathbf{w}}(\mathbf{x}) = \text{sign}\left(w_1 x_1 + w_2 x_2 + \cdots + w_n x_n + w_0\right)$$
$$= \begin{cases} +1, & \text{if } w_1 x_1 + w_2 x_2 + \cdots + w_n x_n + w_0 > 0 \\ -1, & \text{if } w_1 x_1 + w_2 x_2 + \cdots + w_n x_n + w_0 \leq 0 \end{cases}$$

  where $\mathbf{w} = (w_0, w_1, \ldots, w_n)$
- $w_1, \ldots, w_n \in \mathbb{R}$ are called **weights**, and $w_0 \in \mathbb{R}$ is called the **bias**
- Each choice of $\mathbf{w} = (w_0, w_1, \ldots, w_n)$ gives a different hypothesis $h_{\mathbf{w}} \in \mathcal{H}$
- $\mathcal{H}$ is the set obtained when all the possible values of the weights are considered

$$\mathcal{H} = \{h_{\mathbf{w}} : w_0, w_1, \ldots, w_n \in \mathbb{R}\}$$

Interpreting the perceptron score

- The quantity $w_1 x_1 + w_2 x_2 + \cdots + w_n x_n + w_0$ is a **weighted score** of the features $\mathbf{x}$

Example

In the credit approval problem, the perceptron score is:

$$w_1 \cdot \text{income} + w_2 \cdot \text{years employed} + w_3 \cdot \text{outstanding debt} + w_0$$

- Assuming $w_1, w_2 > 0$ and $w_3 < 0$, one could argue that:
  - Large positive score $\Rightarrow$ confidently approve
  - Large negative score $\Rightarrow$ confidently deny

  with w_0 determining where the approve/deny threshold is

- The sign$(\cdot)$ function turns this score into a final decision, $+1$ or -1

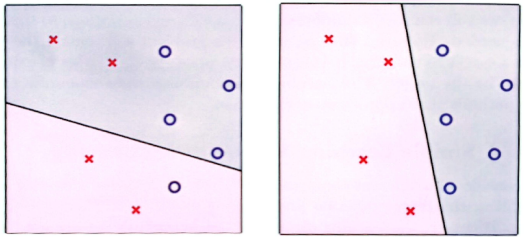
Geometric interpretation

- With 2 features, the boundary

$$w_1 x_1 + w_2 x_2 + w_0 = 0$$

  is a **line** in the plane

- It splits the plane into an $h = +1$ region and an $h = -1$ region
- In n dimensions, this boundary is a **hyperplane**



(a) Misclassified data (b) Perfectly classified data

Source: Abu-Mostafa, Y., Magdon-Ismail, M., & Lin, H.-T. (2012). *Learning from Data*. AMLBook. Figure 1.3.

The perceptron learning problem

- **Goal**: find weights $w_1, \ldots, w_n$ and bias w_0 so that *every* training example is correctly classified
- A training example $(\mathbf{x}_i, y_i)$ is said to be **misclassified** when

$$\text{sign}\left(w_1 x_{i1} + \cdots + w_n x_{in} + w_0\right) \neq y_i$$

- We need an algorithm that searches for such weights automatically
- The idea is simple and **iterative**:
  - Start anywhere (*e.g.*, all weights zero)
  - Repeatedly find a training point that is currently misclassified, and **correct** the weights to fix it
  - Stop when no misclassified point remains
- Let us now write this precisely

The Perceptron Learning Algorithm (PLA)

Assuming that the dataset is linearly separable

PLA

1. **Initialise**: $w_1(0) = \cdots = w_n(0) = 0,\ w_0(0) = 0$
2. For each iteration $t = 1, 2, \ldots$:
   - Find any **misclassified** training example $(\mathbf{x}_i, y_i)$
   - **Update**, for $j = 1, \ldots, n$:

$$w_j(t) \leftarrow w_j(t-1) + y_i\, x_{ij} \qquad\qquad w_0(t) \leftarrow w_0(t-1) + y_i$$

3. **Repeat** step 2 until no misclassified example remains

Numerical example

Example

Consider two training points:

$$\mathbf{x}_1 = (1, 1), \quad y_1 = +1 \qquad\qquad \mathbf{x}_2 = (-1, -1), \quad y_2 = -1$$

- We will run the PLA on this tiny dataset, by hand, step by step
- Initialise: $w_1 = 0$, $w_2 = 0$, $w_0 = 0$
- Score for $\mathbf{x}_1 = (1, 1)$:

$$w_1 x_{1,1} + w_2 x_{1,2} + w_0 = 0 \cdot 1 + 0 \cdot 1 + 0 = 0$$

- A score of 0 counts as -1, hence this point is misclassified
- We will update the weights using this point

Numerical example

- Update using $(\mathbf{x}_1, y_1) = ((1,1), +1)$:

$$w_1 \leftarrow 0 + (+1) \cdot 1 = 1$$
$$w_2 \leftarrow 0 + (+1) \cdot 1 = 1$$
$$w_0 \leftarrow 0 + (+1) = 1$$

- New weights: $w_1 = 1$, $w_2 = 1$, $w_0 = 1$
- Score for $\mathbf{x}_2 = (-1, -1)$ with the updated weights:

$$1 \cdot (-1) + 1 \cdot (-1) + 1 = -1$$

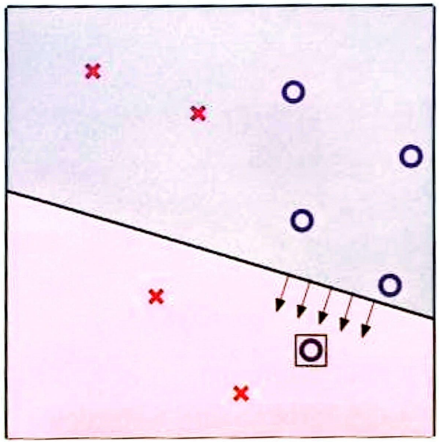
- $\text{sign}(-1) = -1 = y_2$, which is **correctly classified**
- Re-check $\mathbf{x}_1 = (1,1)$ with the updated weights:

$$1 \cdot 1 + 1 \cdot 1 + 1 = 3 > 0$$

- $\text{sign}(3) = +1 = y_1$, which is **correctly classified**
- Both training points are now correctly classified and the PLA **stops** after a single update

PLA: geometric intuition

- Each update **nudges** the boundary toward correctly classifying the misclassified point
- Over several iterations, the boundary **rotates**
- Eventually, every training point ends up on the correct side, even if a given step undoes the correction of previous steps



Source: Abu-Mostafa, Y., Magdon-Ismail, M., & Lin, H.-T. (2012). *Learning from Data*. AMLBook. p. 7.

Convergence of the PLA

Key result

If the training data is **linearly separable**, that is, if some choice of weights and bias classifies *every* training point correctly, then the PLA is guaranteed to **stop** after finitely many updates, and returns such a solution.

- This is a remarkable fact: a simple, greedy, correcting one-mistake-at-a-time algorithm is guaranteed to succeed **if** a solution exists
- What if **no** perfect linear boundary exists?
  - Then the data is **not linearly separable**
  - In this case, the PLA never finds a hypothesis with zero mistakes: it will loop **forever**
  - We will see, later in the course, algorithms that handle this case

Have we solved the problem?

- Suppose the PLA converges for the credit approval problem: $\hat{h}$ has **zero error** on the training data
- But the bank will use $\hat{h}$ on *future* applicants, not seen during training

Feasibility of learning from data

Does performing well on the training data imply performing well on new data? We are back at the **problem of induction**!

Generalisation and Probability

In-sample error

- **In-sample error**: the fraction of **training** examples misclassified by $\hat{h}$

$$E_{\text{in}}(\hat{h}) = \frac{1}{N} \sum_{i=1}^{N} \mathbb{1}\{\hat{h}(\mathbf{x}_i) \neq y_i\}$$

- This is something we can **compute exactly** since we have access to the training dataset
- The PLA, by design, drives $E_{\text{in}}(\hat{h})$ toward zero
- But can we somehow assess how the learned hypothesis $\hat{h}$ will perform on unseen data?

Does the training data tell us anything outside it?

- Let $\mathcal{X} = \{0,1\}^3$: three binary inputs, so there are $2^3 = 8$ possible inputs in total
- We are given a training set $\mathcal{D}$ of **five** labelled examples, leaving **three** inputs unseen
- The five training examples, with $y_n = f(\mathbf{x}_n)$ in which white $=$ 0 and black $= 1$
- The learning algorithm finds $\hat{h}$ that agrees with all five examples on $\mathcal{D}$
- The three remaining inputs are **outside** $\mathcal{D}$ and their labels are unknown to us

x_1	x_2	x_3	y_n
0	0	0	○
0	0	1	●
0	1	0	●
0	1	1	○
1	0	0	●
1	0	1	?
1	1	0	?
1	1	1	?

How many target functions are consistent with $\mathcal{D}$?

- Since f is **unknown**, any function that agrees with $\mathcal{D}$ on the five training points is a valid candidate for the true f

- The three unseen inputs each have a binary output: $\circ$ or $\bullet$, giving $2^3 = \mathbf{8}$ possible functions $f_1, \ldots, f_8$, all consistent with $\mathcal{D}$

$\mathbf{x}$	y	g	f_1	f_2	f_3	f_4	f_5	f_6	f_7	f_8
0 0 0	$\circ$	$\circ$	$\circ$	$\circ$	$\circ$	$\circ$	$\circ$	$\circ$	$\circ$	$\circ$
0 0 1	$\bullet$	$\bullet$	$\bullet$	$\bullet$	$\bullet$	$\bullet$	$\bullet$	$\bullet$	$\bullet$	$\bullet$
0 1 0	$\bullet$	$\bullet$	$\bullet$	$\bullet$	$\bullet$	$\bullet$	$\bullet$	$\bullet$	$\bullet$	$\bullet$
0 1 1	$\circ$	$\circ$	$\circ$	$\circ$	$\circ$	$\circ$	$\circ$	$\circ$	$\circ$	$\circ$
1 0 0	$\bullet$	$\bullet$	$\bullet$	$\bullet$	$\bullet$	$\bullet$	$\bullet$	$\bullet$	$\bullet$	$\bullet$
1 0 1		?	$\circ$	$\circ$	$\circ$	$\circ$	$\bullet$	$\bullet$	$\bullet$	$\bullet$
1 1 0		?	$\circ$	$\circ$	$\bullet$	$\bullet$	$\circ$	$\circ$	$\bullet$	$\bullet$
1 1 1		?	$\circ$	$\bullet$	$\circ$	$\bullet$	$\circ$	$\bullet$	$\circ$	$\bullet$

Source: Abu-Mostafa, Y., Magdon-Ismail, M., & Lin, H.-T. (2012). *Learning from Data*. AMLBook. p. 17.

Learning seems infeasible

- $\hat{h}$ is chosen to match $\mathcal{D}$, but on the three unseen points, $\hat{h}$ must make **some** prediction
- Whatever $\hat{h}$ predicts on each of those three points, exactly **4 out of 8** candidate functions will agree with it, and the other 4 will disagree
- The data provides **no basis** for preferring one prediction over another on unseen inputs
- This result is not specific to Boolean functions: as long as f is unknown, $\mathcal{D}$ cannot rule out **any** pattern of values for f outside $\mathcal{D}$
- Does this mean learning is impossible?

Learning is feasible

Fortunately, **no**, but escaping this conclusion requires a fundamentally different point of view, a **probabilistic** one

Probability within the learning framework

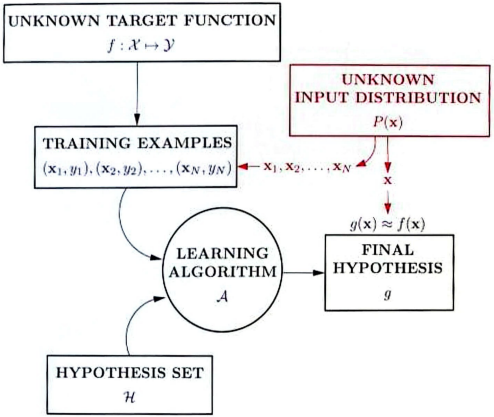


Figure 1.9: Probability added to the basic learning setup

Source: Abu-Mostafa, Y., Magdon-Ismail, M., & Lin, H.-T. (2012). *Learning from Data*. AMLBook. Figure 1.9.

Induction in a probabilistic sense

- We cannot **guarantee** $E_{\text{in}}(\hat{h}) \approx E_{\text{out}}(\hat{h}) =$ Error of $\hat{h}$ on data out of the sample
- But we **can** ask: how **probable** is it that $E_{\text{in}}(\hat{h})$ and $E_{\text{out}}(\hat{h})$ are far apart?
- This is a question that **probability theory** is equipped to answer and answering it, rigorously, is the subject of **Statistical Learning Theory**
- Without probability and statistics, we can *train* a model, but we cannot say anything rigorous about whether it will work on new data
- Statistics will let us:
  - **Quantify** the gap between $E_{\text{in}}(\hat{h})$ and $E_{\text{out}}(\hat{h})$
  - Determine **how much data** is needed for reliable generalisation
  - Understand how the **complexity** of $\mathcal{H}$ controls this gap

Lecture highlights

- **Science**: hypotheses confronted with evidence, kept only while not falsified; **induction** goes from data to a general rule, exactly what ML does
- Three types of ML: **supervised**, **unsupervised**, **reinforcement**
- **Supervised ML framework**: unknown f, training data, hypothesis set $\mathcal{H}$, algorithm $\mathcal{A}$, learned hypothesis $\hat{h} \approx f$
- **Perceptron**: linear threshold classifier; PLA converges whenever the data is linearly separable
- Does small $E_{\text{in}}(\hat{h})$ imply small $E_{\text{out}}(\hat{h})$? This is **generalisation**, and answering it needs **probability and statistics**
- The next lecture will review the **probability and statistics** concepts we will need throughout this course. With those tools in hand, we return to the generalisation question and answer it **precisely**

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